

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import statsmodels.api as sm
import statsmodels.formula.api as smf
import scipy.stats as stats
```

```
# grab the data from online, Excel is fine
url = 'https://www.qogdata.pol.gu.se/data/qog_bas_cs_jan24.xlsx'
df = pd.read_excel(url)
```

```
df.head()
```

	ccode	cname	ccode_qog	cname_qog	ccodealp	ccodecow	version	ajr_settmort	atop_ally	atop_number
0	4	Afghanistan	4	Afghanistan	AFG	700.0	QoGBasCSjan24	4.540098	1.0	1.0
1	8	Albania	8	Albania	ALB	339.0	QoGBasCSjan24	NaN	1.0	8.0
2	12	Algeria	12	Algeria	DZA	615.0	QoGBasCSjan24	4.359270	1.0	9.0
3	20	Andorra	20	Andorra	AND	232.0	QoGBasCSjan24	NaN	1.0	2.0
4	24	Angola	24	Angola	AGO	540.0	QoGBasCSjan24	5.634789	1.0	8.0

5 rows x 337 columns

1-- Run a simple bivariate regression, and interpret your results. (Did the results fit your expectations? Why? Why not?)

Dependent Variable: World Happiness Report (whr_hap). You can see the distribution of it here.

```
df[['whr_hap']].describe()
```

	whr_hap	
count	149.000000	
mean	5.487879	
std	1.130867	
min	2.436034	
25%	4.778489	
50%	5.474300	
75%	6.259810	
max	7.889350	

Independent Variable: GDP per capita (wdi_gdpcappppcon2017), represents the average national income in 2017. Higher values indicate higher average national income and a stronger overall economy, and lower values indicate lower average national income and a weaker overall economy. This is the description of the GDP per capita variable.

```
df[['wdi_gdpcappppcon2017']].describe()
```

wdi_gdpcappppcon2017



count	183.000000
mean	19532.769577
std	20194.334395
min	711.355286
25%	4860.802490
50%	13175.521484
75%	27024.709961
max	111751.312500



I expect that there is a relationship where countries with higher GDP per capita tend to have higher average happiness levels.

(Notice the `subset=df['undp_hdi'].notna()` element below, which restricts the analysis to countries that are not missing data on the HDI variable, which I am going to add in the next analysis, and need to be not missing.)

```
happiness_gdp = smf.ols(formula = 'whr_hap~wdi_gdpcappppcon2017', data = df, subset=df['undp_hdi'].notna()).fit()
print (happiness_gdp.summary())
```

OLS Regression Results

Dep. Variable:	whr_hap	R-squared:	0.569			
Model:	OLS	Adj. R-squared:	0.566			
Method:	Least Squares	F-statistic:	188.5			
Date:	Sun, 26 Oct 2025	Prob (F-statistic):	6.78e-28			
Time:	15:41:02	Log-Likelihood:	-159.62			
No. Observations:	145	AIC:	323.2			
Df Residuals:	143	BIC:	329.2			
Df Model:	1					
Covariance Type:	nonrobust					
=====						
	coef	std err	t	P> t	[0.025	0.975]

Intercept	4.6702	0.086	54.045	0.000	4.499	4.841
wdi_gdpcappppcon2017	4.074e-05	2.97e-06	13.731	0.000	3.49e-05	4.66e-05
=====						
Omnibus:	23.198	Durbin-Watson:	1.909			
Prob(Omnibus):	0.000	Jarque-Bera (JB):	29.712			
Skew:	-0.942	Prob(JB):	3.53e-07			
Kurtosis:	4.170	Cond. No.	4.14e+04			
=====						

Notes:

- [1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
- [2] The condition number is large, 4.14e+04. This might indicate that there are strong multicollinearity or other numerical problems.

Coefficient for GDP per capita is 0.0000404, which indicates that for each additional dollar in GDP per capita, the average national happiness score increases by 0.00004 points. In other words, an increase of 10,000 USD in GDP per capita is associated with about a 0.4 point rise in the average happiness score (on a 0-10 scale). The model is statistically significant with $p < 0.001$. The R-squared of 0.56 indicates a strong association.

Indeed, the regression suggests a strong and positive relationship where wealthier economies tend to have higher levels of reported happiness.

2-- Add an additional variable that might mediate or partly "explain" the initial association from that simple regression above -- and explain your results. Did it work out? Yes? No?

But economic prosperity may not directly translate into happiness, so I include the Human Development Index (`undp_hdi`), which ranges from 0 (lowest human development) to 1 (highest human development). I would expect that, with this new variable added, it will absorb some of the effect of GDP per capita, because countries with higher income levels often achieve better education and health outcomes, both of which contribute directly to higher life satisfaction and happiness.

```
df[['undp_hdi']].describe()
```

	undp_hdi	
count	189.000000	
mean	0.720349	
std	0.149286	
min	0.386000	
25%	0.604000	
50%	0.736000	
75%	0.830000	
max	0.959000	

How do my two independent variables correlate? They correlate meaningfully (+0.80). This indicates that countries with higher GDP per capita also tend to have higher Human Development Index scores.

```
df_filtered = df[['wdi_gdpcappppcon2017', 'undp_hdi']].dropna()
stats.pearsonr(df_filtered['wdi_gdpcappppcon2017'], df_filtered['undp_hdi'])
```

```
PearsonRResult(statistic=np.float64(0.7975525954023963), pvalue=np.float64(3.808310662243824e-41))
```

```
happiness_gdp_hdi = smf.ols(formula='whr_hap ~ wdi_gdpcappppcon2017 + undp_hdi', data=df).fit()
print(happiness_gdp_hdi.summary())
```

```

=====
                        OLS Regression Results
=====
Dep. Variable:          whr_hap      R-squared:                0.669
Model:                  OLS          Adj. R-squared:            0.664
Method:                 Least Squares   F-statistic:              143.4
Date:                   Sun, 26 Oct 2025   Prob (F-statistic):       8.27e-35
Time:                   15:41:02         Log-Likelihood:           -140.46
No. Observations:        145             AIC:                     286.9
Df Residuals:            142             BIC:                     295.8
Df Model:                2
Covariance Type:        nonrobust
=====

```

	coef	std err	t	P> t	[0.025	0.975]
Intercept	2.3018	0.369	6.235	0.000	1.572	3.032
wdi_gdpcappppcon2017	1.664e-05	4.51e-06	3.693	0.000	7.73e-06	2.56e-05
undp_hdi	3.9210	0.598	6.555	0.000	2.739	5.103

```

=====
Omnibus:                 22.249    Durbin-Watson:              1.782
Prob(Omnibus):           0.000    Jarque-Bera (JB):         37.914
Skew:                    -0.740    Prob(JB):                 5.85e-09
Kurtosis:                 5.021    Cond. No.:                3.81e+05
=====

```

Notes:

- [1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
- [2] The condition number is large, 3.81e+05. This might indicate that there are strong multicollinearity or other numerical problems.

GDP per capita (wdi_gdpcappppcon2017) has a coefficient of 1.66e-05, indicating that for each additional dollar in GDP per capita, the average national happiness score increases by approximately 0.000016 points (where it goes up to 10), net of human development.

The Human Development Index (undp_hdi), which measures overall human well-being on a scale from 0 (lowest) to 1 (highest), has a positive coefficient of 3.92. This suggests that, controlling for GDP per capita, countries with higher levels of human development tend to have substantially higher happiness scores. This result is statistically significant ($p < 0.001$).

The adjusted R-squared is 0.664, which is notably higher than in the earlier model with GDP alone (0.566). This indicates that the addition of human development improves the explanatory power of the model. Happiness is influenced not only by income but also by many aspects of well-being such as health and education.

The biggest finding here is that the Human Development Index (HDI) does not fully mediate the relationship between GDP and happiness. Even after accounting for human development, higher income levels continue to have an independent positive effect on happiness, though the effect size is much smaller. This suggests that both economic prosperity and human development contribute to happiness, but the latter plays a more dominant role.

The results are consistent with expectations and show that when the HDI is added to the model, it takes up part of the effect of GDP on happiness. GDP is still positive and statistically significant, but its coefficient becomes smaller. This means that some of the impact of income on happiness likely works through improvements in such as education, health etc. that HDI represents.

3-- More on extreme combinations. Find the top 5 entities that are ranked at the top on one variable and ranked at the bottom on another variable. Interpret your results.

I am going to look at countries that are relatively high in GDP per capita, but do not appear to have correspondingly high levels of happiness among their citizens.

```
df['gdp_rank'] = df['wdi_gdpcapppcon2017'].rank(ascending=False)
df['happiness_rank'] = df['whr_hap'].rank(ascending=False)

extreme_comb = df[['cname', 'gdp_rank', 'happiness_rank']].copy()
extreme_comb['rank_difference'] = extreme_comb['happiness_rank'] - extreme_comb['gdp_rank']
extreme_sorted = extreme_comb.sort_values(by='rank_difference', ascending=False).head(5)

extreme_sorted
```

	cname	gdp_rank	happiness_rank	rank_difference	
94	Lebanon	83.0	148.0	65.0	
177	Turkey	45.0	107.0	62.0	
19	Botswana	86.0	143.0	57.0	
90	Korea (the Republic of)	24.0	63.0	39.0	
78	Iran (Islamic Republic of)	77.0	106.0	29.0	

后续步骤: [使用 extreme_sorted 生成代码](#) [New interactive sheet](#)

These results highlight a significant difference in these countries between their level of economic prosperity and the subjective well-being of their citizens. This could suggest a potential disconnect or inefficiency in translating material wealth into overall life satisfaction. It is interesting that several of these countries such as Lebanon, Turkey, and Iran have faced political instability or social tensions, which may help explain why relatively high income levels do not correspond with equally high happiness outcomes.